

PIC16F877A-based Temperature Monitoring System

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Temperature monitoring and control is important in industrial environments. Sensors are widely used for measurement of temperature. Usually, a temperature sensor converts the temperature into an equivalent voltage output. IC LM35 is such a sensor. Here we describe a simple temperature measurement and display system based on LM35 sensor and PIC16F877A microcontroller. The temperature in degrees Celsius is displayed on a 16x2 LCD.

Fig. 1 shows the functional block diagram of the PIC16F877A-based temperature monitoring system. The key features of this system are:

1. Continuous monitoring of temperature with 1-second update interval (which can be varied in the program)
2. Temperature measurement using LM35 precision integrated-circuit sensor
3. Precise analogue-to-digital conversion using in-built 10-bit analogue-to-digital converter (ADC) of PIC16F877A microcontroller

Hardware description

Fig. 2. shows the circuit of the temperature monitoring system. The circuit mainly consists of the LM35 temperature sensor, PIC16F877A microcontroller and HD44780 controller based 16×2 LCD.

The output of the sensor is fed to

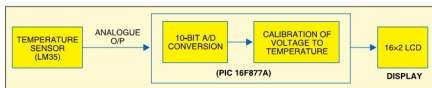


Fig. 1: Block diagram of PIC16F877A-based temperature monitoring system



the internal ADC of the microcontroller. Pin 2 of the microcontroller (RA0/AN0) is channel-1 of the internal ADC. The analogue voltage output of the sensor is converted into its equivalent digital value by the ADC and then its equivalent degree Celsius value is calculated by the software. The calculated temperature value is displayed on the LCD.

LM35 sensor. Fig. 3 shows the pin configuration of LM35. It is a precision integrated-circuit centigrade temperature sensor whose output voltage is linearly proportional to the Celsius (Centigrade) temperature. The LM35 thus has an advantage over linear temperature sensors calibrated in degree Kelvin, as the user is not required to subtract a large constant voltage from its output to obtain convenient Centigrade scaling. For each degree Celsius change in temperature, the sensor output changes by 10 mV.

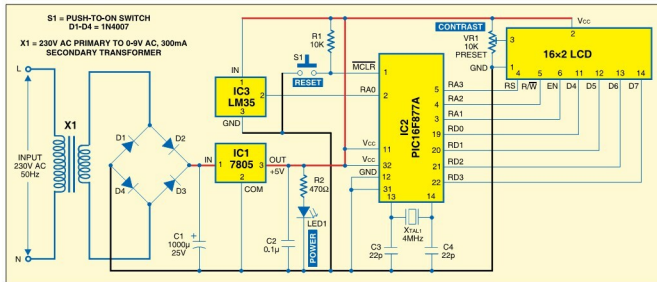


Fig. 2: Circuit of PIC16F877A-based temperature monitoring system